Section 5.4 - Additional Curve Sketching

Graphing Strategy:

- **Step 1:** Analyze $f(x)$
  - Find the domain of f.
  - Find the intercepts.
  - Find asymptotes.
- **Step 2:** Analyze $f'(x)$: Create sign chart for $f'(x)$ and determine intervals of increasing/decreasing and local extrema.
- **Step 3:** Anlayze $f''(x)$: Create sign chart for $f''(x)$ and determine intervals of concave up, concave down, and inflection points.
- **Step 4:** Sketch the graph of f. ~~using all of the above information.~~ Plot additional points as needed.

create a combined # line to determine the shape of the graph on each interval and then

Example 1: Use the graphing strategy to sketch a graph of $f(x) = \dfrac{2+x}{3-x}$.

Step 1:

$\boxed{\text{Domain: } x \neq 3}$

y-int (x=0)

$f(0) = \frac{2+0}{3-0} = \frac{2}{3}$

$\boxed{(0, 2/3)}$

x-int (y=0)

$0 = \frac{2+x}{3-x}$

$0 = 2+x$

$-2 = x$ $\boxed{(-2,0)}$

$f(3) = \frac{\text{nonzero\#}}{0}$ $\boxed{x=3 \text{ is a VA}}$

$\lim_{x\to\pm\infty} \frac{2+x}{3-x} = \lim_{x\to\pm\infty} \frac{x}{-x} = \lim_{x\to\pm\infty} -1 = -1$

$\boxed{y=-1 \text{ is HA}}$

Step 2:

$f'(x) = \frac{(3-x)(1)-(2+x)(-1)}{(3-x)^2}$

$= \frac{3-\cancel{x}+2+\cancel{x}}{(3-x)^2}$

$= \frac{5}{(3-x)^2} = 5(3-x)^{-2}$

f'(x) is never zero but undefined when x = 3

		x=0		3		x=4
f(x)			↗	⋮	↗	
f'(x)		+		⋮	+	

$\boxed{\text{Increasing } (-\infty,3)\cup(3,\infty) \text{ NO Local Extrema}}$

Step 3:

$f''(x) = 5 \cdot -2(3-x)^{-3} \cdot (-1)$

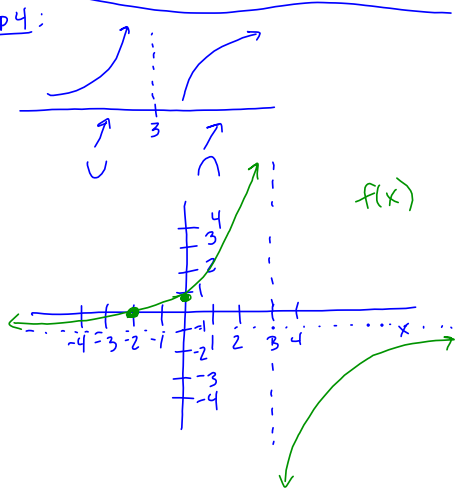
$= \frac{10}{(3-x)^3}$

f''(x) is never zero but undefined when x = 3

		x=0		3		x=4
f(x)			∪	⋮	∩	
f''(x)		+		⋮	−	

$\boxed{\text{Concave Up } (-\infty,3) \text{ Concave Down } (3,\infty) \text{ No Inflection Point}}$

Step 4:



Example 2: Use the graphing strategy to sketch a graph of $f(x) = (x-2)e^x$.

*DO THIS for clicker question on Wed. *

use a table to convince yourself for HA.

Step 1:

Domain: $\mathbb{R}$

y-int

$f(0) = (-2)e^0 = -2$

$\boxed{(0,-2)}$

x-int

$0 = (x-2)e^x$

$x-2=0$

$x=2$

$\boxed{(2,0)}$

$\boxed{\text{No vertical asymptotes}}$

$\lim_{x\to\infty} (x-2)e^x = \infty$

$\lim_{x\to-\infty} (x-2)e^x = 0$

create table to convince

$\boxed{y=0 \text{ is HA but only in } -\infty \text{ direction}}$

Step 2:

$f'(x) = (x-2)e^x + e^x(1)$

$= e^x(x-2+1)$

$= e^x(x-1)$

$f'(x) = 0$ when $x = 1$

		$x=0$		$x=2$
$f(x)$	↘		↗	
$f'(x)$	$-$		$+$	

$\boxed{\text{Increasing } (1,\infty) \quad \text{Decreasing } (-\infty,1) \quad \text{Local min } (1,-e)}$

Step 3:

$f''(x) = e^x(1) + (x-1)(e^x)$

$= e^x(1+x-1)$

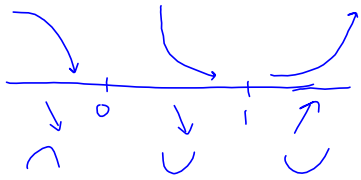
$= xe^x$

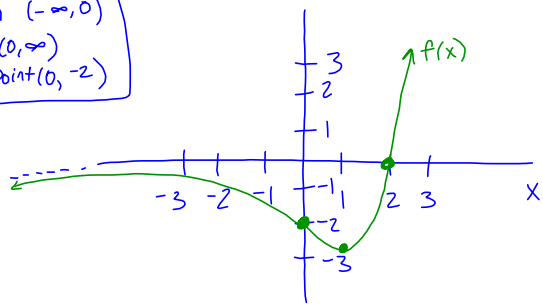
$f''(x) = 0$ when $x=0$

		$x=-1$		0		$x=1$
$f(x)$	∩				∪	
$f''(x)$		$-$			$+$	

$\boxed{\text{Concave Down } (-\infty,0) \quad \text{Concave Up } (0,\infty) \quad \text{Inflection Point } (0,-2)}$

Step 4:



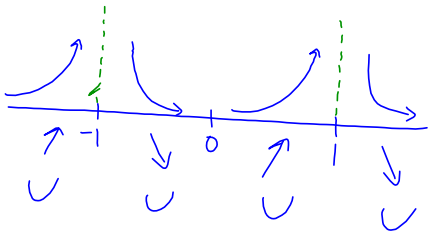


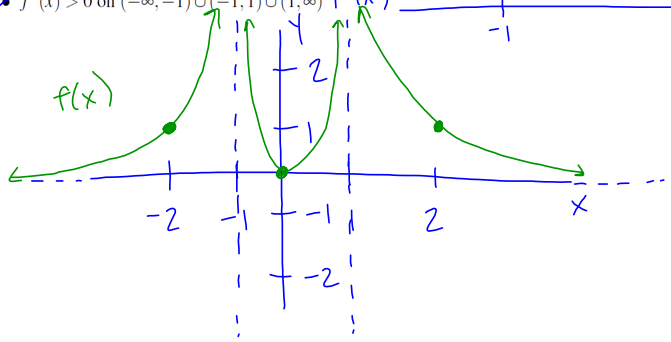
Example 3: Use the given information to sketch the graph of f.

- ✓ Domain: $(-\infty,-1)\cup(-1,1)\cup(1,\infty)$
- ✓ Asymptotes: $x=-1$, $x=1$, $y=0$
- ✓ $f(-2)=1$, $f(0)=0$, $f(2)=1$
- ✓ $f'(x)>0$ on $(-\infty,-1)\cup(0,1)$
- ✓ $f'(x)<0$ on $(-1,0)\cup(1,\infty)$
- ✓ $f''(x)>0$ on $(-\infty,-1)\cup(-1,1)\cup(1,\infty)$

		-1		0		1	
$f(x)$	↗		↘		↗		↘
$f'(x)$	$+$		$-$		$+$		$-$

		-1		1	
$f(x)$	∪		∪		∪
$f''(x)$	$+$		$+$		$+$





Section 5.4 Highly Suggested Homework Problems: 1-35(odd), 41, 47, 49